Ideation Phase
Brainstorm & Idea Prioritization Template

Date	22 February 2026
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Project Name	Heart Disease Analysis
Maximum Marks	4 Marks

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Project Focus

Heart disease is one of the leading global health issues. Through interactive Tableau dashboards, we aim to identify key risk factors and patterns that influence heart disease occurrence using our dataset.

Selected Problem Statement

How can interactive Tableau visualizations help identify key risk factors, patterns, and trends related to heart disease based on lifestyle, medical conditions, and demographic factors?

Step-2: Brainstorm, Idea Listing and Grouping

A) Data Preparation (Completed)

- Removed duplicates from the dataset.
- Checked and handled missing or invalid values.
- Converted 0/1 values into Yes/No labels.
- Standardized column names for clarity.
- Ensured correct data types (Measures & Dimensions in Tableau).
- Converted categorical fields (Sex, AgeCategory, Race, etc.) into proper labels.
- Created calculated fields:
  - BMI Category (Underweight / Normal / Overweight / Obese)
  - Risk Level (Based on BMI, Smoking, AlcoholDrinking, and AgeCategory)
  - Sleep Category (Low / Normal / High Sleep)

B) Key Visualization Ideas

- KPI Dashboard: % Heart Disease, Average BMI, Average Sleep Time, % Smokers, % Diabetic.
- Bar Charts: Heart Disease by AgeCategory, Sex, Smoking, AlcoholDrinking, Diabetic, PhysicalActivity, GenHealth.
- Health Condition Analysis: Stroke, KidneyDisease, Asthma, DiffWalking vs Heart Disease.
- Line Charts: BMI vs PhysicalHealth, SleepTime vs MentalHealth.
- Heatmap: Risk Factor Combination (BMI Category + AgeCategory).
- Pie Charts: Heart Disease by Race, Heart Disease vs PhysicalActivity.
- Interactive Filters: AgeCategory, Sex, Race, Diabetic, PhysicalActivity.
- Risk Segmentation Dashboard using Risk Level.
- Story Points for final presentation.

C) Dashboard Grouping

1. Risk Factor Dashboard
2. Health Condition Dashboard
3. Lifestyle & Wellness Dashboard

Step-3: Idea Prioritization

Prioritization Criteria: Impact, Feasibility, Effort, and Clarity.

Idea / Feature	Impact	Feasibility	Effort	Priority
KPI Dashboard	High	High	Low	High
Heart Disease by AgeCategory	High	High	Low	High
Smoking vs Heart Disease	High	High	Low	High
Diabetic vs Heart Disease	High	High	Low	High
Line Chart Analysis	Medium	High	Low	Medium

Heatmap (BMI + AgeCategory)	**High**	**Medium**	**Medium**	**Medium**
Pie Chart Analysis	**Medium**	**High**	**Low**	**Medium**
Advanced Analytics / Clustering	**Low**	**Medium**	**High**	**Low**